The Response of the Ideal Ocean Planet to the Tidal Force

[redacted]

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Abstract

Tides have been a critical focus in Earth science and astronomy due to their universal importance. While previous studies offer quantitative descriptions of tidal forces, the coherence of their logical frameworks requires further examination. This study addresses this gap by reevaluating existing deductions and presenting a comprehensive, logically consistent derivation of tidal forces. I quantitatively analyze fluid responses to tidal forces, detailing transitions from initial to steady states, and qualitatively explore tidal lagging. Utilizing Newton's equilibrium tide model, I simplify dynamic problems into static ones and apply energy perspectives to resolve these issues. Our results enhance the understanding of tidal force logic, refining theoretical approaches and offering broader insights into tidal phenomena.

Key words: tidal force, equilibrium tide, tidal lagging

1 Introduction

Tide is a quite common but important natural phenomenon, for its function of transporting material between the ocean and the land, remodeling the physiognomy of the coast line from the geographical perspective. The distribution of tidal force causes the difference of resultant force on different geological elements, which influences the deformation of the rock, resulting the global geological movement. It is also attached importance to in the celestial mechanics, affecting the dissipation of mechanical energy and the evolution of the binary star systems, like the tidal locking and tidal tearing. The progress and effects of the tidal locking of the Moon is still the frontier problem in the field of Planetary Science.

I have preliminarily learned the deduction of tidal force in *University Physics* during the second semester of freshman year, and further study this topic in *General Astronomy* and *Theoretical Mechanics* in first semester of sophomore year. However, the response of the ocean to the tidal force has not been discussed, which means the mechanism of the generation and the evolution of tide is not clear. Previous research on this topic also has some flaws. Most articles just give the result of so-called tidal ellipse qualitatively without any deduction. In Ref.1, the reasonable explanation for some approximate treatment is lack. In Ref.5, the deduction of the so-called tidal ellipse is not so convincing for its absence of crucial logical chain.

In this article, detailed deduction of the tidal force and the response of the ocean to the tidal force is discussed based on the ideal model after further research and personal thinking.

2 Context

2.1 Tidal Force

On the basis of the formula of tidal force taught in class, we use the ideal model below to further discuss the tidal force. Consider a planet with mass m and radius R and a mass point M, whose distance to the COM of the planet O is d, as shown in Fig 1. We can get the gravitational potential energy as the function of r, the distance between the mass point and an arbitrary mass point on the surface of the planet, which can then be substituted by θ.

As for the coordinate system, we set O as the origin, and both rectangular coordinate system and cylindrical coordinate system will be applied.

$$r^2 = R^2 + d^2 - 2Rdcos\theta \tag{1}$$

$$V(r) = -\frac{Gm}{r} = -\frac{Gm}{(R^2 + d^2 - 2Rdcos\theta)^{-\frac{1}{2}}} \tag{2}$$

Given that $d \gg R$, we use Taylor expansion on $V(\theta)$,

$$\begin{aligned}
&(R^2 + d^2 - 2Rdcos\theta)^{-\frac{1}{2}} \\
=&d^{-1}[1 + (\frac{R}{d})^2 - 2\frac{R}{d}cos\theta]^{-\frac{1}{2}} \\
=&d^{-1}\{1 - \frac{1}{2}[(\frac{R}{d})^2 - 2\frac{R}{d}cos\theta] + \frac{3}{8}[(\frac{R}{d})^2 - 2\frac{R}{d}cos\theta]^2 + \cdots\cdots\} \\
=&d^{-3}[d^2 + Rdcos\theta + \frac{R^2}{2}(3cos^2\theta - 1) + o(R^2)]
\end{aligned}$$

so that

$$V(\theta) = -\frac{Gm}{d}[1 + \frac{R}{d}cos\theta + \frac{R^2}{2d^2}(-1 + 3cos^2\theta) + o((\frac{R}{d})^2)]. \tag{3}$$

Let

$$V_0 = -\frac{Gm}{d} = Const, \tag{4}$$

$$V_1 = -Gm\frac{R}{d^2}cos\theta = -Gm\frac{z}{d^2}, \tag{5}$$

where $z = Rcos\theta$, and we set $\overrightarrow{Om}$ as the positive direction of z-axis,

$$V_2 = -Gm\frac{R^2}{2d^3}(-1 + 3cos^2\theta). \tag{6}$$

For $\mathbf{g} = -\nabla V$, in the rectangular coordinate system,

$$\mathbf{g_1} = -\nabla V_1 = \frac{Gm}{d^2}\mathbf{k} \tag{7}$$

and we can easily figure out that $\mathbf{g_1}$ is the common $\mathbf{g}$ when we consider the gravitational potential between two mass points, and is the gravitational potential at the position of O in the gravitational field of the mass point M, as well.

And in the cylindrical coordinate system,

$$\mathbf{g_2} = -\nabla V_2 = -(\mathbf{e_r}\frac{\partial}{\partial r} + \mathbf{e}_\theta\frac{\partial}{r\partial\theta})V_2 = \mathbf{g_r} + \mathbf{g}_\theta, \tag{8}$$

$$where \quad \mathbf{g_r} = -\mathbf{e_r}\frac{\partial V_2}{\partial r} = Gm\frac{R}{d^3}(3cos^2\theta - 1)\mathbf{e_r}, \tag{9}$$

$$and \quad \mathbf{g}_\theta = -\mathbf{e}_\theta\frac{\partial V_2}{r\partial\theta} = -Gm\frac{R}{d^3}(3cos\theta sin\theta)\mathbf{e}_\theta. \tag{10}$$

which is the tidal force.

2.2 Tidal Ellipse

Then we will discuss the function of the tidal force on the planet. In order to simplify the analysis, we examine this ideal planet, which is covered by ocean with the same depth at any place before being influenced by the tidal force. When the gravitation of the mass point is taken into consideration, we use such non-inertial reference frame, where the planet and the mass point are relatively rest. For the system is steady, the inertial centrifugal force is balanced by the gravitational force on the COM of the planet O, which is indeed the resultant force of $\Delta m g_1$ on an object on the planet.

Before further discussion, we calculate the order of magnitudes of the gravitation F_G, the inertial centrifugal force F_c caused by the auto-rotation of planet and the tidal force F_t on the surface of the planet. Take the Earth-Moon system as an example. Earth represents the planet, while Moon is regarded as the mass point.

The constant has the following value:

$$\omega = 7.292{\times}10^{-5} rad/s, M = 5.972{\times}10^{24} kg, m = 7.342{\times}10^{22} kg,$$
$$d = 3.844{\times}10^{8} m, a = 6.371{\times}10^{6} m, G = 6.674{\times}10^{-11} N{\cdot}m^2/kg^2,$$

, where d is the average distance between Earth and Moon, and a is the average radius of Earth. We get

$$\begin{aligned}\frac{F_G}{\Delta m} &= \frac{GM}{a^2} \approx 10^1, \\ \frac{F_c}{\Delta m} &= \omega^2 a \approx 10^{-4}, \\ \frac{F_t}{\Delta m} &= \frac{Gma}{d^3} \approx 10^{-7}, \end{aligned} \tag{11}$$

which indicates that in the radial direction, the tidal force and the inertial centrifugal force can be ignored compared with the gravitation. By contrast, in the angle direction, initially the object is only influenced by the tidal force $\mathbf{g}_\theta$, which causes the ocean to flow, resulting the increase amount of water and the increase of depth consequently at some particular places. And that is actually the formation of tide. The distribution of water depth according to angle θ leads to the change of the direction of the water pressure. When the ocean reaches the steady state, in the angle direction, the component of the tidal force is balanced by the component of the pressure.

But the examination of the water pressure is much more complex, for it requires the master of fluid mechanics. However, we can turn to the perspective of energy, ignoring the details of the redistribution of force. When the ocean reaches the steady state, suppose in a small part of region there exists the inequality of potential energy, and then the ocean will flow continuously, for the gradient of the potential energy is not zero, which is contrary to the definition of the steady state. As a result, we can conclude that the surface of the ocean must be the equi-potential plane.

Consider the potential energy of the tidal force $\Delta m V_2$, and the potential energy of the gravitation of the planet E_G. Inspired by the approach of dealing with small oscillation in *Theoretical Mechanics*, choose the surface of the ocean at the initial state as the potential energy zero. The deviation from the potential energy zero is h. For simplification, $\Delta mgh = \Delta m\frac{GM}{a^2}h$ is noted as E_G. At the steady state, we have

$$\Delta m V_2 + E_G = Const, \tag{12}$$

and we get

$$-G(\Delta m)m\frac{R^2}{2d^3}(-1+3cos^2\theta)+(\Delta m)\frac{GM}{a^2}h = Const, \tag{13}$$

where $R = a + h$. In (13),the second order is dominant, for the remarkable difference in the order of magnitudes of the tidal force and the gravitation, as shown in (11). Thus, we can use a to represent R,

$$-\frac{ma^2}{2d^3}(-1+3cos^2\theta)+\frac{M}{a^2}h = Const, \tag{14}$$

$$h = \frac{ma^4}{2Md^3}(-1+3cos^2\theta)+Const. \tag{15}$$

For the conservation of mass during the distribution of water, the integral of h at the volume element should be zero, which is

$$\iiint h\mathrm{d}R\mathrm{d}\theta\mathrm{d}\phi = 0 \tag{16}$$

Thus, $Const = 0$.

We get

$$R = a + h = a(1+\frac{ma^3}{2Md^3}(-1+3cos^2\theta)). \tag{17}$$

In the rectangular coordinate system where $x = Rcos\theta, y = Rsin\theta$, we have

$$(x^2+y^2)\sqrt{x^2+y^2} = Ax^2+By^2, \tag{18}$$

where

$$A = a+\frac{ma^4}{Md^3}, B = a-\frac{ma^4}{2Md^3}.$$

For $\frac{ma^4}{Md^3} \approx 10^{-1}$ and $a \approx 10^6$, it is hard to see the difference in the x-axis direction and y-axis direction, as shown in the Fig.2.

2.3 Tidal Lagging

Finally, I will discuss the mechanism of tidal lagging. First, the planet has auto-rotation. Second, the response of the solid earth is anelasticity. The long axis of the tidal ellipse and the line connecting the mass point of both planets, like Earth and Moon is not overlapped. The angle between the two lines is called delay angle.

The delay angle can be treated as constant. When considering only the effects of auto-rotation, the ocean is influenced by frictional and gravitational forces. The frictional force between rock and water is independent of the angle, whereas the Moon's gravitational force on Earth is directly proportional to it. As the angle increases, the system reaches a stable (quasi-static) state where the delay angle remains unchanged for a period. This simplification is applied in Reference 4. However, the author of Reference 4 does not identify the appropriate scenario for considering tidal effects. Reference 8 provides a detailed discussion of this model, suggesting it only approximately reflects real conditions over short periods.

Suppose the auto-rotation of the planet is faster than the revolution of the planet, the delay angle leads to the moment of the tidal force on the planet. The moment is reversely parallel to the direction of the angle velocity of auto-rotation, causing the planet to rotate slower, which is the tidal lagging. The angular momentum remains constant all the time. The angular momentum of the planet auto-rotation is partly shifted to the revolution of the satellite. At the same time, the friction between the rock and water on the planet leads to energy dissipation. The moment forces the auto-rotation and the revolution to evolve to the state of sync, leading to the tidal locking.

For the binary system, the total angular momentum L is constant.

$$L = \frac{2}{5}Mr_M^2\omega_M + \frac{2}{5}mr_m\omega_m + \frac{Mm}{M+m}\sqrt{G(M+m)a(1-e^2)} = Const, \quad (19)$$

where e is the eccentricity ratio of satellite's orbit.

For $M \gg m, r_M \gg r_m, e \ll 1, \omega_M = \frac{2\pi}{P_M}$, where P_M is the period of the planet's auto-rotation, we get

$$L = \frac{4\pi}{5} M \frac{r_M^2}{P_M} + M\sqrt{GMa} = Const. \tag{20}$$

Take the derivation of time, we get

$$0 = -\frac{4\pi}{5} M \frac{r_M^2}{P_M^2} \frac{\mathrm{d}P_M}{\mathrm{d}t} + \frac{1}{2} M \sqrt{\frac{GM}{a}} \frac{\mathrm{d}a}{\mathrm{t}}. \tag{21}$$

So

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \frac{8\pi r_M^2}{5MP_M^2} \sqrt{\frac{Ma}{G}} \frac{\mathrm{d}P_M}{\mathrm{d}t}. \tag{22}$$

However, recent study (Ref.14) has shown that the total dissipation rates for most of the past 252 Million years of the Moon were far below present levels. It indicates that $\frac{\mathrm{d}a}{\mathrm{d}t}$ is not constant, implying that the inversion of the Moon's orbit simply based the current stage is impossible. Further study should consider more constrains like paleogeographic or climate configurations.

3 Conclusion

In the non-inertial reference frame, the inertial centrifugal force is parallel anywhere, while the gravitation force is the function of angle. Such deviation results in the tidal force. In the model of steady state, the deformation of the ocean on the ideal ocean could be deduced. The shape of the ocean in the steady state is approximately an ellipse, which is the reason for the so-called tidal ellipse. The tidal ellipse, though tough, can efficiently help explain and predict the scale of tide.

However, more factors should be taken into consideration and more complex models should be discussed to better explain the motion of tide. For example, the model of steady state neglects the horizontal motion of the water, bound to fail to explain the waves. Moreover, the response of the solid earth to the tidal force is intriguing, because it may contribute to the geological movement and the evolution of the Earth-Moon System. But more delicate investigation requires the knowledge of elastic mechanics and celestial mechanics. As a result, further discussion cannot be give in this article of trial.

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Appendix

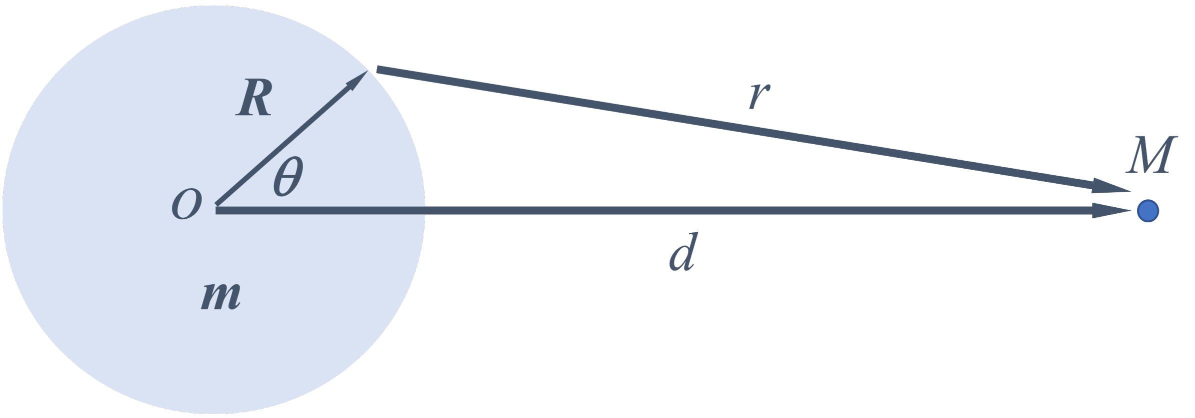

Figure 1: The Model

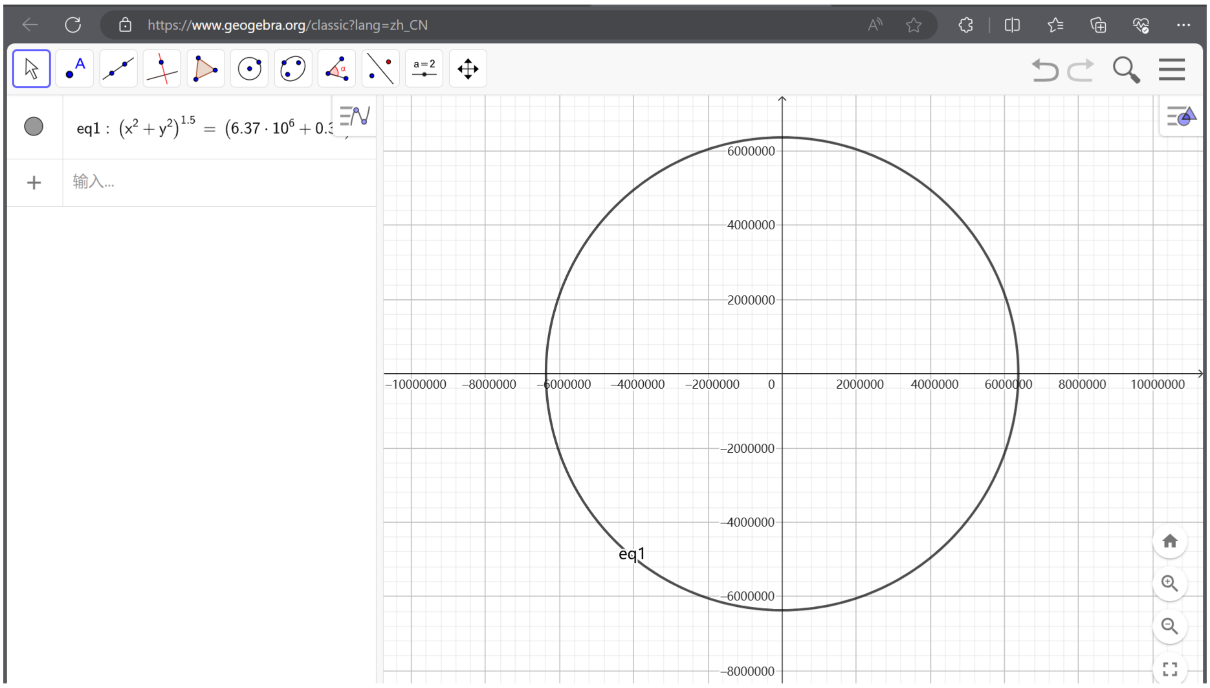

Figure 2: The Tidal Ellipse

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